

Age, Growth, and Feeding Ecology of *Halobatrachus didactylus* in Ria Formosa

Luana Gomes (87878), Maria Arredondo (87877), Eleonora Savino (89606), Brittany Beaudry (87875)

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Abstract

Ria Formosa is a biologically diverse ecosystem critical for maintaining local marine biodiversity and supporting fisheries. This study investigates the biology and ecology of *Halobatrachus didactylus*, focusing on age and growth, reproduction, and diet. 394 specimens were sampled and analyzed encompassing otolith-based age determination, gonadal and liver indices, and dietary assessment through stomach content analysis. Growth patterns revealed sex-specific differences, with females reaching larger asymptotic lengths but males exhibiting faster growth rates. Reproductive analysis showed significant associations between sex and maturity stages, with gonadosomatic and hepatosomatic indices reflecting energy allocation across reproductive phases. Diet analysis highlighted a strong preference for crustaceans, particularly brachyuran crabs, supplemented by mollusks and fish. These findings enhance our understanding of the ecological role of *H. didactylus* as a benthic predator and bioindicator, providing a foundation for conservation strategies in coastal lagoon systems.

1. Introduction

Coastal lagoons are among the most productive and biologically diverse ecosystems on Earth. Acting as transitional zones between terrestrial and marine environments, they provide unique habitats that sustain intricate food webs and facilitate nutrient cycling on both regional and global scales (Pérez-Ruzafa et al., 2024; Andrisoa et al., 2024). Temperate and subtropical lagoons, in particular, are biodiversity hotspots characterized by high productivity and structural complexity (Kennish & Paerl, 2010). These systems serve crucial ecological roles as nurseries for pelagic species and habitats for sedentary fish, which deliver essential ecosystem services (Molina et al., 2023; Pérez-Ruzafa et al., 2024).

The Ria Formosa, located along Portugal's southern coast, exemplifies the ecological importance of coastal lagoons and stands out as one of Europe's premier biodiversity hotspots.

Spanning 55 km and covering a catchment area of approximately 864 km², this lagoon complex hosts a diverse array of habitats that support a remarkable variety of marine organisms (Newton et al., 2013; Duarte et al., 2008). Fish species in the Ria Formosa can be broadly categorized as sedentary, migratory, or accidental. Among these, sedentary species like the toadfish (*Halobatrachus didactylus*) are particularly significant due to their role in structuring benthic communities and their potential as bioindicators of ecosystem health (Palazón-Fernández et al., 2001). Because sedentary fish remain within a single habitat throughout their life cycle, their health and population dynamics often reflect localized environmental conditions, making them valuable indicators of ecosystem quality (Whitfield & Elliott, 2002).

H. didactylus, a demersal teleost, is the sole representative of the family Batrachoididae along the continental Portuguese coast. This subtropical species is most abundant in southern Portugal, with populations on the western coast primarily confined to estuaries and sheltered areas (Cotter et al., 2013). Preferring sandy or muddy substrates in shallow waters (less than 50 m), *H. didactylus* inhabits estuaries and lagoons, where it contributes to food web dynamics by regulating invertebrate populations (Costa et al., 2000). Its diet, primarily composed of benthic invertebrates, underscores its ecological role and utility as a bioindicator of prey availability and ecosystem health (Costa et al., 2000). Additionally, its sedentary nature and strong site fidelity make it particularly sensitive to environmental changes, enhancing its suitability for monitoring habitat integrity and water quality. Gonadal analyses of this species provide insights into reproductive maturity, which is essential for understanding population sustainability and developing conservation strategies (Flores et al., 2019). The reproductive biology of *H. didactylus*, including changes in gonadal morphology throughout its annual cycle, further highlights its potential as a model species for studying reproductive processes (Modesto & Canário, 2003).

To address the limited knowledge of *H. didactylus* in the Ria Formosa, this study aims to explore its biological and ecological characteristics within the context of this lagoon. Specifically, the research investigates three key aspects: age and growth, reproduction, and diet. Age and growth are examined through otolith analysis, a widely used method for estimating the age and growth patterns of fish by analyzing calcified structures (Schöne & Surge, 2012). Reproductive dynamics are assessed through gonadal analysis to determine maturity and reproductive potential, providing insights into population sustainability. Finally, dietary analysis, through stomach content examination, evaluates trophic interactions and resource use, which reflect broader ecosystem health and inform conservation strategies.

(Zacharia, 2011). Together, these approaches aim to provide a holistic understanding of the toadfish's role within the Ria Formosa ecosystem.

While previous studies have explored the histological and biochemical aspects of *H. didactylus* reproduction (Palazón-Fernández, 2001), research focused specifically on populations in the Ria Formosa is scarce. This study seeks to fill this gap by characterizing how the toadfish interacts with the unique environmental conditions of the lagoon. By investigating its growth, reproductive patterns, and feeding ecology, this research paper addresses fundamental questions about the species' ecological role and adaptation to this habitat. Ultimately, the findings aim to enhance our understanding of *H. didactylus* as a bioindicator species and support efforts to monitor and preserve the ecological integrity of the Ria Formosa.

2. Materials and Methods

2.1 Sampling and Specimen Processing

A total of 394 toadfish (*H. didactylus*) were randomly captured in the Ria Formosa lagoon in July 2024 using bottom trawl fishing gear equipped with a fine mesh size (<1 cm). After capture, only 106 specimens were retained and immediately frozen onboard to preserve their condition and transported to the laboratory for analysis. In the laboratory, the specimens were thawed to room temperature prior to processing, which included morphometric measurements, dissection, and subsequent analyses.

2.2 Measurements and Dissection Procedures in the Laboratory

Morphometric and Weight Measurements:

In the laboratory, each specimen was processed to obtain key biological measurements and samples for analysis: total length (TL) was measured to the nearest millimeter using an ichthyometer, along side body weight (BW) using a precision electronic balance (Antoine et al., 2019).

Dissection Procedures:

The abdominal cavity of each fish was carefully incised to remove internal organs for further analysis: gonads and liver were removed and weighed separately (± 0.01 g) to determine reproductive and liver conditions. Stomach contents were also extracted and preserved in 70 % ethanol for dietary analysis (Hyslop, 1980). The largest of the three otolith pairs, the sagittae,

were carefully extracted, rinsed with tap water to remove debris, and air-dried (Panfili & Troadec, 2002). These were used for age determination.

Classification:

Specimens were categorized by sex (male, female, or undetermined) based on gonadal inspection. Maturity stages were assigned using macroscopic criteria adapted from Garcia-Diaz (1997). Fish lacking identifiable gonads were classified as juveniles, while those with gonads but insufficient data for maturity stage assignment were classified as uncertain.

Maturity Stage Determination:

Maturity stages (MS) were determined using a two-step process: first, macroscopically, according to the five-point scale described (Table 1); second, microscopically, by examining gonadal tissue to confirm and refine MS classification. Histological identification of the various maturity stages was determined according to the development of the ovary and testis.

Table 1. Maturity stages and macroscopic determination criteria.

Maturity Stage	Description
I Immature	Gonads absent or too small to determine sex. Less than 1/4 of the body cavity; whitish or pale cream. Juvenile phase.
II Developing	Gonads 1/4–1/2 of body cavity; whitish/pale cream. Promaturation phase, smooth and greyish, ready to mate.
III Pre-Spawning	Gonads > 3/4 of cavity; pink parts with opaque white oocytes visible.
IV Spawning	Swollen and reddish/pinkish gonads with visible oocytes, likely after first spawning event.
V Recovering	Gonads < 1/4 of cavity; flaccid, pale or greyish. Post-spawning stage; visible opaque oocytes or bloody appearance.
VI Uncertain	No details provided or uncertain classification.

Condition Indices:

The following indices were calculated to assess reproductive and liver conditions:

- Gonadosomatic index (GSI): $\frac{\text{Gonad Weight (g)}}{\text{Total Weight (g)}} \times 100$

- Hepatosomatic index (HSI): $\frac{\text{Liver Weight (g)}}{\text{Total Weight (g)}} \times 100$

2.2 Age and Growth Analysis

Otolith Preparation and Reading

Sagittal otoliths were examined under transmitted light against a dark background to identify growth rings (translucent and opaque bands). To enhance contrast, otoliths were halved and lightly burned using a Bunsen burner (Christensen, 1964). Growth rings were counted by trained personnel to estimate fish age (Panfili & Troadec, 2002).

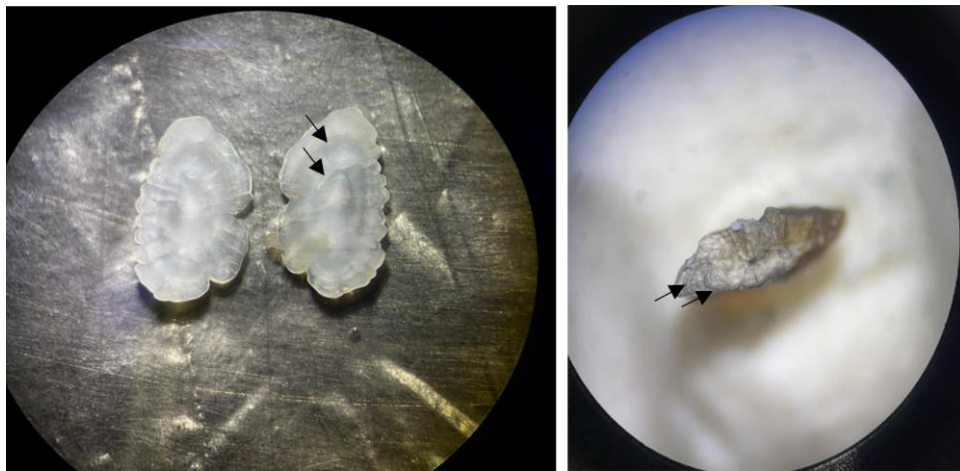


Figure 1. Interpretation of the grown rings (alternating translucent and opaque bands) of sagittal otoliths of *Halobatrachus didactylus*.

Von Bertalanffy Growth Model:

The age-length relationship was modeled using the Von Bertalanffy Growth function (VBFG):

$$L_t = L_{\infty} (1 - e^{-k(t-t_0)}).$$

where L_t is the length at age t , L_{∞} is the predicted asymptotic length, K is the growth coefficient, and t_0 is the theoretical age when fish length is zero.

Parameters were optimized to minimize sum-of-squares errors (SS) and ensure a robust fit using Excel Solver. The VBFG model was applied separately to males and females to assess potential sex-specific differences in growth rates.

This approach was selected due to its ability to account for the biological and environmental factors influencing growth rates, making it a robust tool for analyzing size-at-age data in marine species (Pauly, 1980).

2.3 Reproductive Analysis

Maturity stages were identified through macroscopic examination of the gonads and classified into five categories: Immature, Developing, Pre-Spawning, Spawning, and Recovering. These stages were then correlated with total length and weight, while variations in the gonadosomatic index (GSI) and hepatosomatic index (HSI) were also evaluated.

2.4 Diet Analysis

A subset of 89 stomachs containing prey items were analyzed and identified to the lowest possible taxonomic level, using a microscope, and grouped into the following categories and subcategories: crustaceans (anomura, brachyura, decapod, macrura), echinoderms, fish, mollusks (bivalves and gastropoda), plants, and unclassified (Saldanha, 1997).



Figure 2. Laboratory analysis of the toadfish stomach content samples and classification of the respective taxonomic level.

Diet Indices

Four indices were used to evaluate the relative importance of prey items: the **Percent Number Index (PNI)**, the **Percent Weight Index (PWI)**, and the **Index of Relative Importance (IRI)**.

1. **Percent Number Index (PNI):** The number of individuals of each prey type was expressed as a percentage of the total number of prey items identified in the stomach contents (Hyslop, 1980).
2. **Percent Weight Index (PWI):** The weight of each prey type was calculated as a percentage of the total biomass of prey found in the stomachs (Hyslop, 1980).
3. **Frequency of Occurrence (FO):** The frequency of occurrence of each prey type was determined by calculating the percentage of stomachs containing at least one instance of the prey type (Hyslop, 1980). This was computed as:

$$F = \frac{NSi}{NS\ tot} \times 100$$

4. **Index of Relative Importance (IRI):** This index combines the number, weight, and frequency of occurrence of each prey type (Pinkas et al., 1971). The formula for the IRI is:

$$IRI = (PNI + PWI) \times FO,$$

where N% represents the percentage based on the number of prey items, W% is the percentage based on the weight of prey, and FO% is the frequency of occurrence of the prey type in the sampled stomachs.

2.5 Statistical Analysis

All statistical analyses were performed in R Studio (version 2023.12.1+402), with a significance level set at $p < 0.05$. Figures were created using both Microsoft Excel and R Studio, utilizing the ggplot2 and dplyr packages (Microsoft Corporation, 2024; R Core Team, 2024; Wickham, 2016; Wickham et al., 2023).

To compare the length-frequency distributions between sexes, a Chi-squared test of independence was conducted. Differences in gonad and liver weights between sexes were assessed using Wilcoxon rank-sum tests, as the data were non-normally distributed (Shapiro-Wilk test) but exhibited homogeneous variances (Levene's test) (Pearson, 1900; Wilcoxon, 1945; Shapiro & Wilk, 1965; Levene, 1960). To evaluate gonad weight across maturity stages, pairwise Wilcoxon rank-sum tests were performed, treating maturity stages as an ordered factor (Immature, Developing, Pre-Spawning, Spawning, Recovering, Uncertain), and the Bonferroni correction was applied to adjust for multiple comparisons (Bonferroni, 1936). Differences in liver weight across maturity stages were assessed using a Kruskal-Wallis rank sum test due to the non-normality of the data (Kruskal & Wallis, 1952). Fisher's Exact Test was used to examine the association between sex and maturity (Fisher, 1922).

3. Results

3.1 Age –growth

The sample used for this experiment consists of toadfish of both sexes, male and female, with total lengths ranging between 5 cm and 39 cm. Certain length classes, such as 18 cm, 26 cm, and 31 cm, were more frequently observed (Figure 3), and while both sexes were represented across most length classes, certain classes showed slightly higher counts for one sex. However, statistical analysis revealed no significant difference in the distribution of males and females across total length categories (p -value = 0.2642), suggesting that the sex ratio does not vary significantly with length.

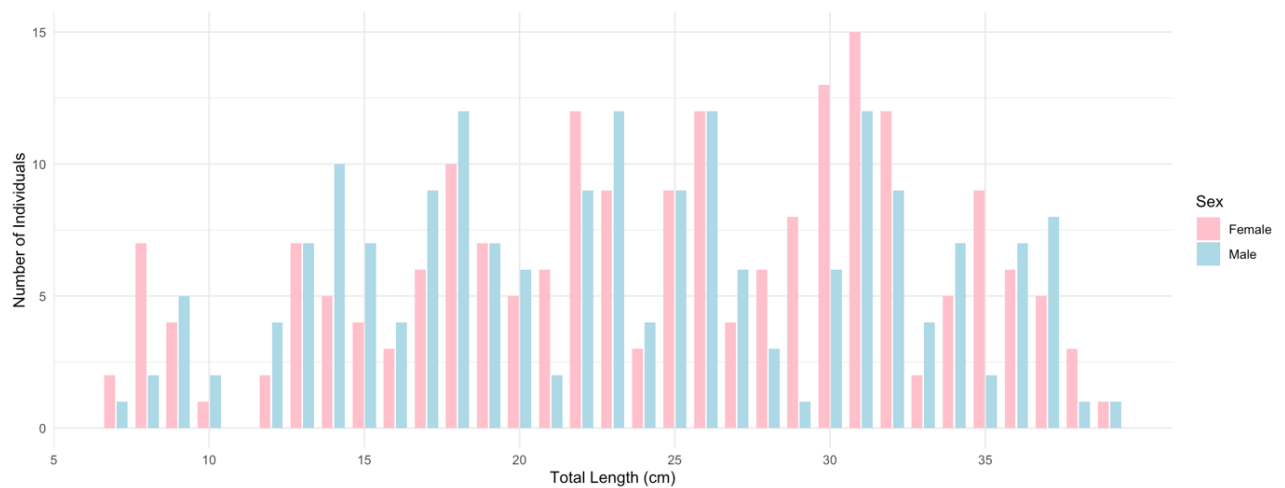


Figure 3. Length-frequency distribution of sampled *H. didactylus* individuals, showing the number of male (blue bars) and female (pink bars) individuals across different total lengths (cm).

Additionally, certain length classes, such as 18 cm, 26 cm, and 31 cm, showed higher numbers of individuals for both sexes, possibly representing dominant size cohorts in the sampled population. In contrast, classes such as 11 cm were underrepresented or absent.

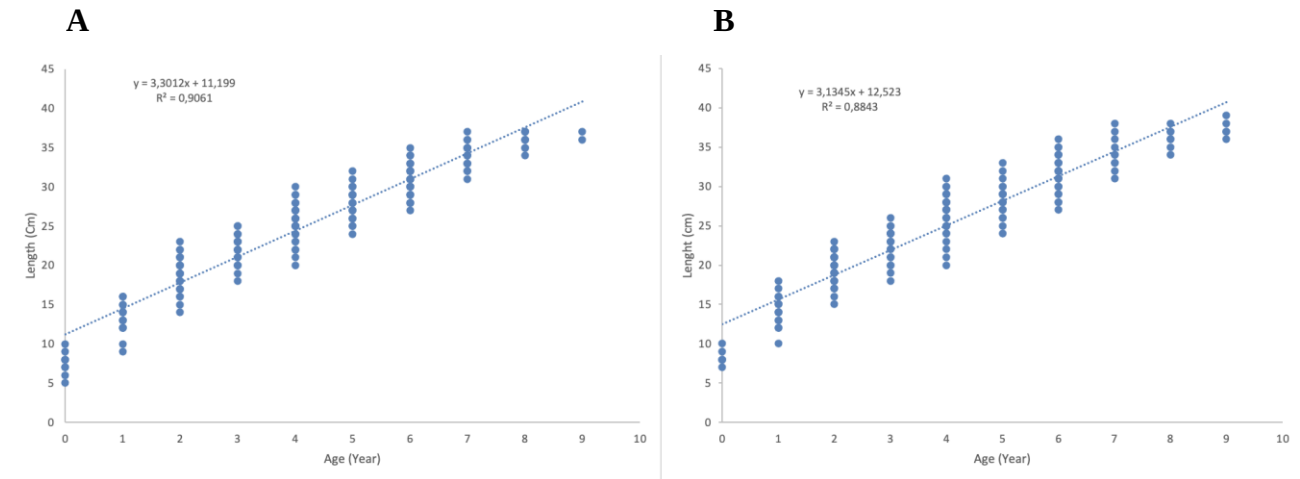


Figure 4. Length-age key correlation for Females (A) and Males (B). It show a strong positive correlations and consistent growth patterns for both sexes, with females growing slightly faster and exhibiting a stronger correlation between age and length compared to males.

The relationship between age and length (Figure 4) shows distinct yet consistent growth patterns for both males and females. In males, the slope of 3.13 indicates an average length increase of 3.13 cm per year, with an intercept of 12.52 cm suggesting a starting length at age zero of approximately 12.52 cm. In females, the slightly steeper slope of 3.30 reflects a faster growth rate, with an average annual length increase of 3.30 cm and a starting length of around 11.20 cm, as indicated by the intercept. The R^2 values demonstrate strong positive correlations for both sexes, with 0.8843 for males and a slightly higher 0.9061 for females, indicating that age explains 88.43% and 90.61% of the variability in length, respectively. These results highlight predictable and robust growth patterns, with females exhibiting slightly faster growth and a stronger correlation between age and length compared to males, potentially reflecting differences in life history strategies.

Age (years)	0		1		2		3		4		5		6		7		8		9		Total N
Length (cm)	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	
<10	0																				0
10-11	0		1	0%																	1
12-13			4	1%																	4
14-15			6	1%	2	0%															8
16-17			3	0%	6	1%															9
18-19			1	0%	14	2%	6	1%													21
20-21					18	3%	12	2%	8	1%											38
22-23					10	2%	15	2%	8	1%											33
24-25							18	3%	20	3%	10	2%									48
26-27							3	0%	28	4%	20	3%	6	1%							57
28-29									28	4%	35	6%	24	4%							87
30-31									12	2%	15	2%	36	6%	7	1%					70
32-33											10	2%	42	7%	21	3%					73
34-35													30	5%	21	3%	16	3%			67
36-37													6	1%	14	2%	32	5%	27	4%	79
38-39															7	1%	8	1%	18	3%	33
Grand Total	0		15	2%	50	8%	54	9%	104	17%	90	14%	144	23%	70	11%	56	9%	45	7%	628

Figure 5. Age-length key results from otolith readings of males specimen.

Age (years)	0		1		2		3		4		5		6		7		8		9		Total N
Length (cm)	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	
<10	0		1	0%																	1
10-11	0		1	0%																	1
12-13			6	1%																	6
14-15			5	1%	4	1%															9
16-17			2	0%	10	1%															12
18-19					18	2%	6	1%													24
20-21					14	2%	18	2%	8	1%											40
22-23					6	1%	27	4%	16	2%											49
24-25							15	2%	32	4%	15	2%									62
26-27									36	5%	30	4%	6	1%							72
28-29									20	3%	40	6%	36	5%							96
30-31									4	1%	20	3%	54	7%	7	1%					85
32-33											5	1%	48	7%	35	5%					88
34-35													18	2%	49	7%	24	3%			91
36-37															21	3%	48	7%	18	2%	87
Grand Total	0		15	2%	52	7%	66	9%	116	16%	110	15%	162	22%	112	15%	72	10%	18	2%	723

Figure 6. Age-length key results from otolith readings of females specimen

The age-length key results for males (Figure 5) and females (Figure 6) highlight distinct growth and population dynamics within the sampled specimens. Both sexes show a clear progression of increasing age with length, with younger individuals (ages 0–2) primarily occupying the smaller size classes (<20 cm), and older individuals (ages 6–9) dominating the larger size classes (>30 cm). Females exhibit slightly greater representation in mid-sized classes (20–27 cm), while males are more frequent in the largest classes (32–37 cm). The majority of individuals for both sexes are aged 6 years, accounting for 23% in males and 22% in females,

underscoring the significance of this cohort. The data suggest that females may have a longer life span, given their stronger representation in the older age groups (7–9 years).

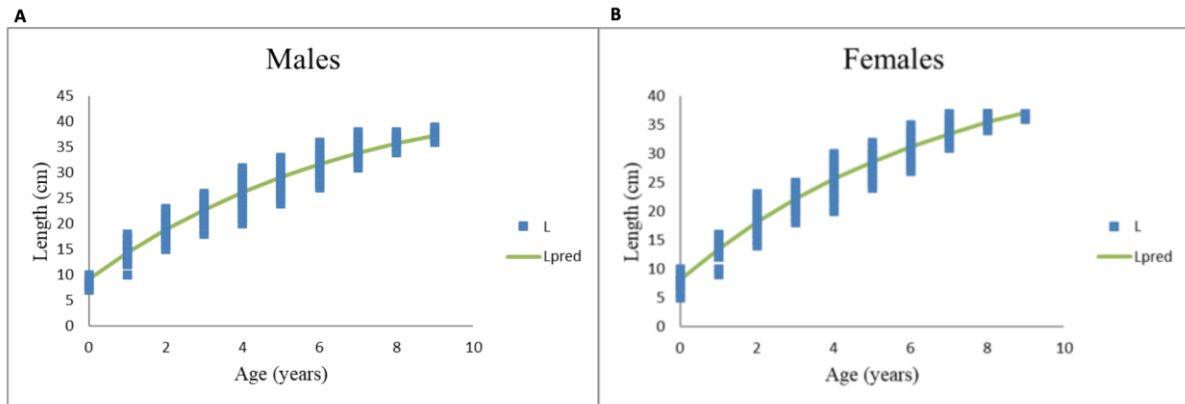


Figure 7. Age-growth curves for males (A) and females (B) with Von-Bertalanffy growth equations.

The von-Bertalanffy growth equation for males (Figure 7.A) is $L_t = 47.601 (1 - e^{-0.147(t+1.439)})$ and for females $L_t = 48.251 (1 - e^{-0.143(t+1.284)})$, revealing distinct growth trends for males and females (Figure 7). The model predicted that females (Figure 7.B) would reach an asymptotic length of $L_\infty = 48.251$, while males would reach $L_\infty = 47.601$ cm. However, the maximum observed sizes in the study were 37 cm for females and 39 cm for males, both of which were smaller than the predicted values. The growth coefficients (K) were slightly higher for males ($K = 0.147$) compared to females ($K = 0.143$), indicating that males tend to reach their asymptotic size more rapidly than females. The t_0 values were negative for both sexes, with females having $t_0 = -1.284$ and males $t_0 = -1.439$. The sum of squares values indicated a reasonable fit of the model to the data, with a slightly better fit for females (731.025) compared to males (774.107).

3.2 Reproduction

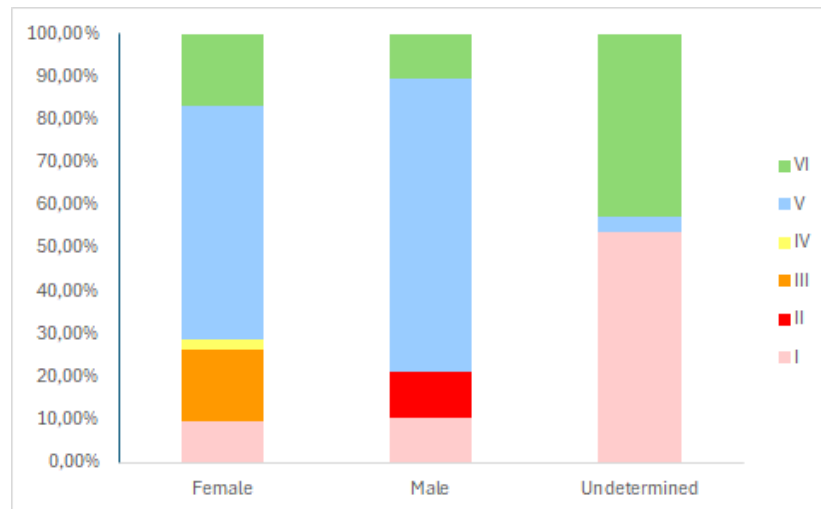


Figure 8. Distribution of the maturity stages (I-Immature; II-Developing; III-Pre-Spawning; IV-Spawning; V-Recovering; VI-Uncertain) for females, males and undertermined.

The macroscopic determination of the maturity stages, as outlined in Table 1, revealed six maturity stages in this study: *Immature*, *Developing*, *Pre-Spawning*, *Spawning*, *Recovering*, and *Uncertain*. The distribution of maturity stages varied significantly across females, males, and undetermined individuals (Figure 8). There was no statistically significant association between sex and maturity ($p = 0.1123$).

Among females, the majority were in stages V (Recovering) and VI (Uncertain), with only a small proportion in stage III (Pre-Spawning) and stages I and II (Immature and Developing, respectively). Males exhibited a similar pattern, with the Recovering stage being the most prevalent, followed by the Uncertain and Developing stages. Only a small fraction of males was in the Pre-Spawning and Immature stages. For undetermined individuals, the distribution was distinct. A significantly higher proportion were in the Immature stage compared to females and males. While the Recovering stage was still present, it was less dominant. The Uncertain stage also made up a notable portion of undetermined individuals, whereas the Spawning stage was almost absent across all groups.

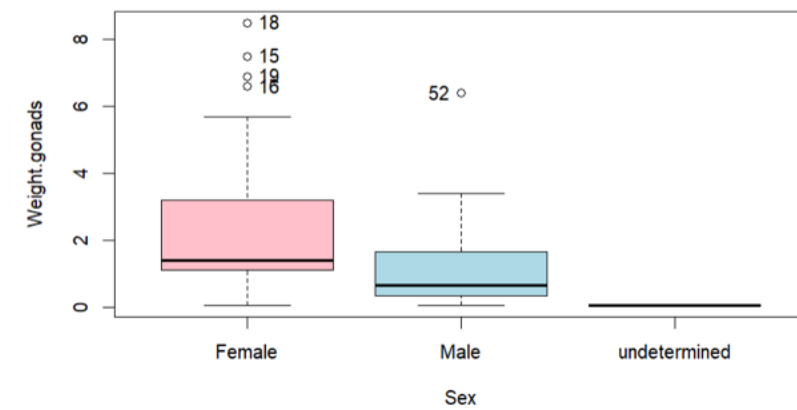


Figure 9. Distribution of gonad weights (g) for females, males, and undetermined individuals. The horizontal line within each box marks the median gonad weight for each group, while the box boundaries indicate the interquartile range (IQR). Outliers are represented as numbered points outside the whiskers of the respective groups.

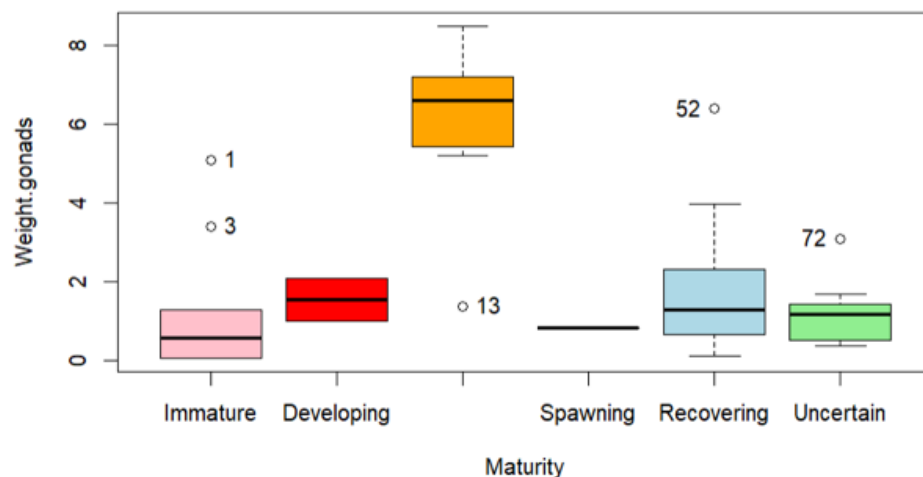


Figure 10. Distribution of gonad weights(g) across different maturity stages: *Immature* (pink), *Developing* (red), *Spawning* (orange), *Recovering* (blue), and *Uncertain* (green). The horizontal line in each box shows the median gonad weight, the box edges represent the interquartile range (IQR), and outliers appear as numbered points outside the whiskers.

The distribution of gonad weights was analyzed in relation to sex and maturity stages (Figures 9 and 10, respectively). Females had higher gonad weights on average compared to males and undetermined individuals, with greater variability and several extreme outliers. Males showed lower gonad weights, with fewer outliers, while the undetermined group had minimal variability and consistently low gonad weights. A Mann-Whitney U test revealed a significant difference between males and females ($W = 321$, $p = 0.0013$).

In terms of maturity stages, spawning individuals exhibited the highest gonad weights, followed by pre-spawning individuals (Figure 10). Developing and Recovering individuals had intermediate gonad weights, and Immature and Uncertain individuals displayed the lowest gonad weights. Pairwise comparisons using the Wilcoxon rank-sum test with Bonferroni correction indicated significant differences between the Immature and Pre-Spawning stages ($p = 0.0062$), Pre-Spawning and Recovering ($p = 0.0053$), and Pre-Spawning and Uncertain ($p = 0.0066$) stages. No significant differences were found between other maturity stage comparisons ($p > 0.05$).

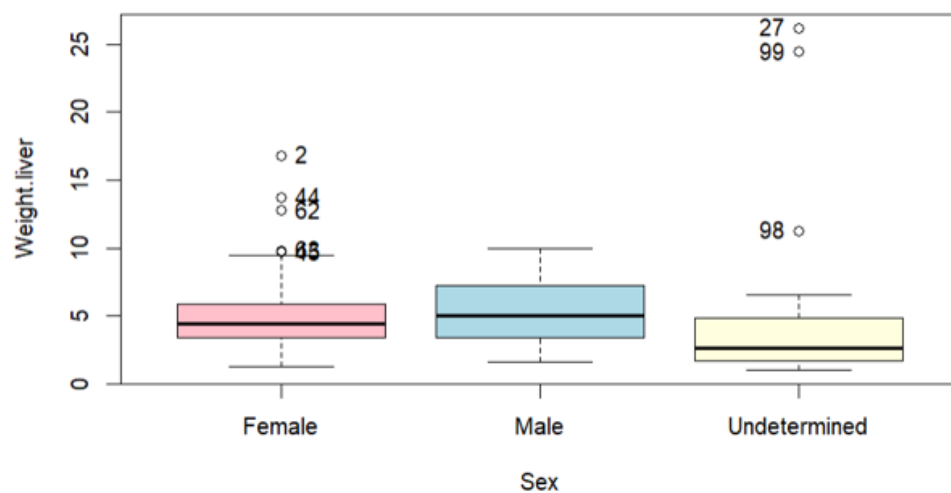


Figure 11. Distribution of liver weights (g) for females, males, and undetermined individuals. The horizontal line within each box marks the median liver weight, while the box boundaries indicate the interquartile range (IQR). Outliers are represented as numbered points outside the whiskers of the respective groups.

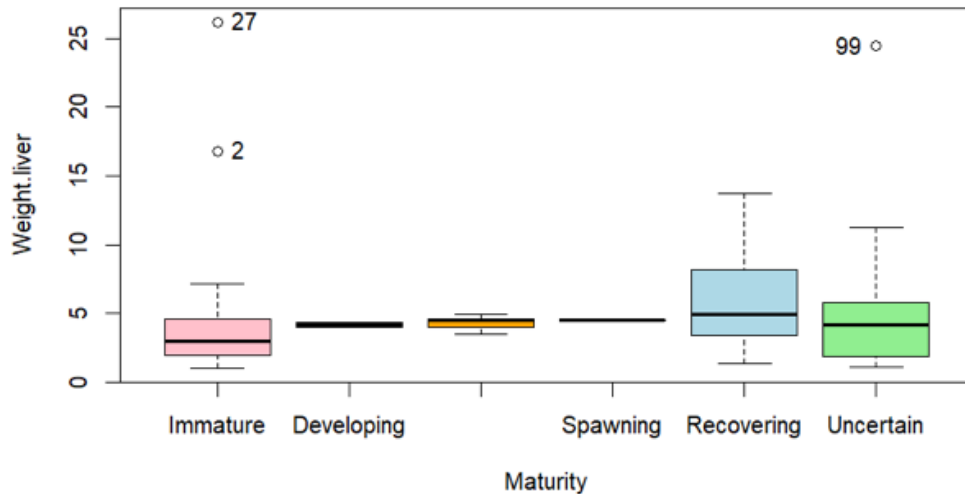


Figure 12. Distribution of liver weights (g) across different maturity stages: *Immature* (pink), *Developing* (red), *Spawning* (orange), *Recovering* (blue), and *Uncertain* (green). The horizontal line in each box shows the median liver weight, the box edges represent the interquartile range (IQR), and outliers appear as numbered points outside the whiskers.

Liver weight was analyzed in relation to sex (Figure 11) and maturity stages (Figure 12). Across sexes (Figure 11), median liver weights were similar for females, males, and undetermined individuals. Males exhibited slightly higher variability, while undetermined individuals had the most extreme outliers. A Wilcoxon rank sum test with continuity correction showed no significant difference in liver weight between sexes ($W = 614.5$, $p = 0.5419$).

In relation to maturity stages (Figure 12), Immature individuals consistently had low liver weights with minimal variability. Developing and Pre-Spawning individuals also showed low liver weights. Recovering individuals had the largest liver weights and the greatest variability. The Uncertain group displayed moderate liver weights with variability similar to the Recovering stage. A Kruskal-Wallis rank sum test revealed no significant differences in liver weight across the six maturity stages ($\chi^2 = 10.046$, $df = 5$, $p = 0.07394$).

A

B

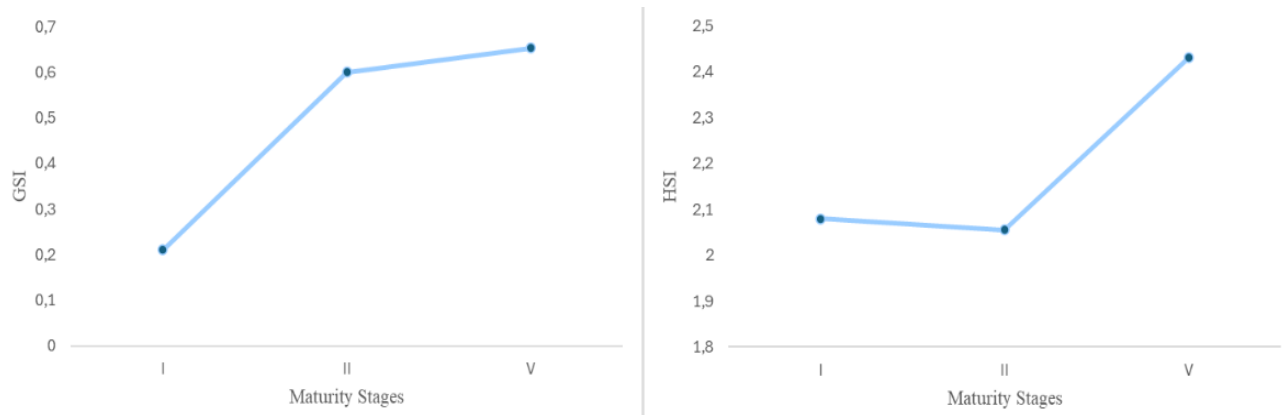


Figure 13. A- Gonadosomatic index for males at different maturity stages; B-Hepatosomatic index for males at different maturity stages.

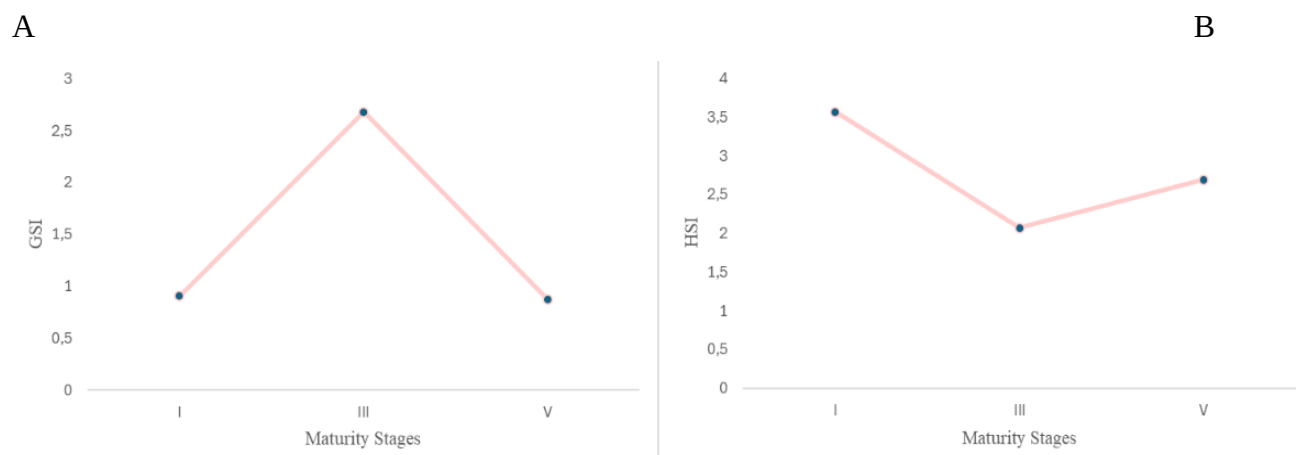


Figure 14. A- Gonadosomatic index for females at different maturity stages; B -Hepatosomatic index for females at different maturity stages.

The **Gonadosomatic Index (GSI)** for males is highest during the Developing stage (II), with a value around 0.65, and lowest during the Immature stage (I), with a value around 0.08 (Figure 13.A).

The **Hepatosomatic Index (HSI)** for males is highest during the Recovering stage (V) and lowest during the Developing stage (II) (Figure 13.B).

The **GSI** for females is highest during the Pre-Spawning stage (III), with a value around 2.7, and lowest during both the Immature stage (I) and the Recovering stage (V), with values around 0.8 (Figure 14, A).

In contrast to males, the **HSI** for females is highest during the Immature stage (I), with a value around 3.5, and lowest during the Pre-Spawning stage (III) (Figure 14, B). Overall, females exhibit higher GSI values compared to males across all stages. The Wilcoxon rank-sum test

revealed a statistically significant difference for GSI values between sexes ($W = 896.5$, $p\text{-value} = 0.0001377$). However, no statistically significant difference was observed for the HSI between sexes ($W = 655$, $p\text{-value} = 0.2717$).

3.3 Diet

Table 2. List of prey items found in the examined stomachs, its total number and weight (g), Percent Number Index (PNI), the Percent Weight Index (PWI), and the Index of Relative Importance (IRI).

Taxonomic groups	Total Number	Weight(g)	PNI %	PWI %	FO %	IRI %
Crustaceans	155	347	49,1	50,2	76,4	75,8
Anomura	13	21,4	4,1	3,1	9,0	0,6
Brachyura	138	323,2	43,7	46,8	62,9	56,9
Decapod	3	1,7	0,9	0,2	3,4	0,0
Macrura	1	0,8	0,3	0,1	1,1	0,0
Echinoderms	23	47,9	7,3	6,9	11,2	1,6
Fish	38	156,3	12,0	22,6	27,0	9,3
Molluscs	90	127,7	28,5	18,5	50,6	23,7
Bivalves	25	29,5	7,9	4,3	15,7	1,9
Gastropoda	65	98,2	20,6	14,2	34,8	12,1
Plant	8	9,8	2,5	1,4	10,1	0,4
Unclassified	2	2,2	0,6	0,3	2,2	0,0

Out of the 106 stomachs examined, 89 contained prey, resulting in a fullness index of approximately 83.96%. For comparison, all identified prey items, and their corresponding indices are detailed in Table 2. Based on these data, pie charts were created to visualize the differences in indices across prey categories.

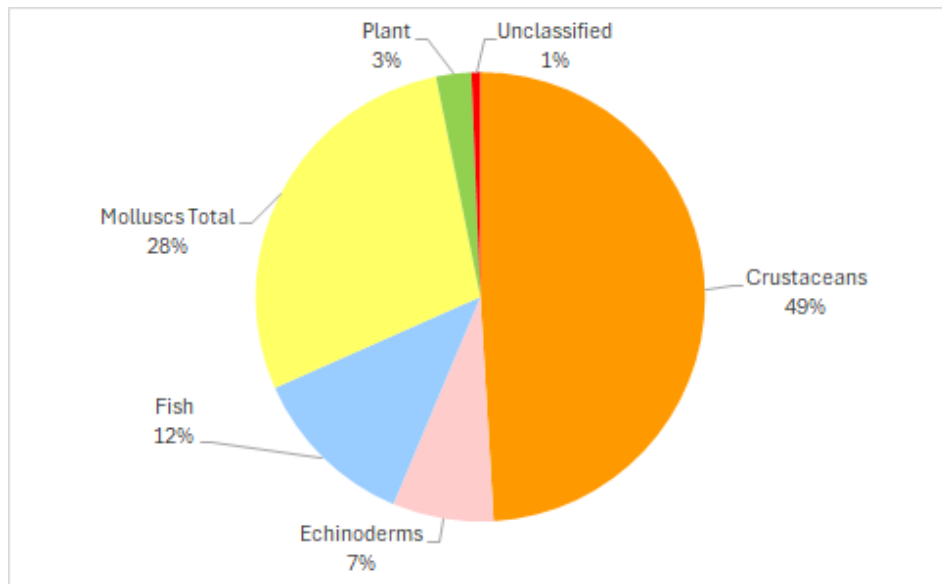


Figura 15. Percentage of the Total Number (N) of prey for the general diet.

By numbers, the most significant group in the diet is crustaceans, representing 49% of the total prey items consumed (Figure 1). This is followed by mollusks, which make up 28%, and fish, contributing 12%. Echinoderms account for 7% of the total number of prey items, while plant matter and unidentified prey items constitute 3% and 1%, respectively

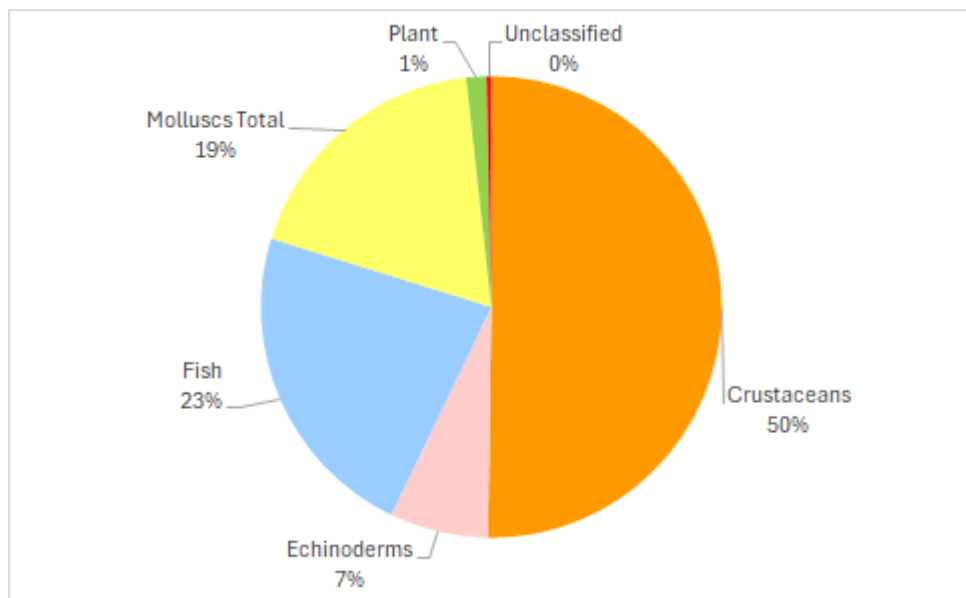


Figura 16. Gravimetric index as a percentage for the general diet.

Crustaceans also dominate the diet by weight, making up 50% of the total prey biomass. Fish follow as the second most important group, contributing 23%, with mollusks close behind at

19%. Echinoderms account for 7%, while plant matter and unidentified prey contribute just 1% and 0%, respectively.

When comparing the two indexes (Figure 15 and 16), the results indicate that crustaceans are the most prominent prey group in both abundance and biomass, while fish and mollusks contribute significantly to the overall diet composition.

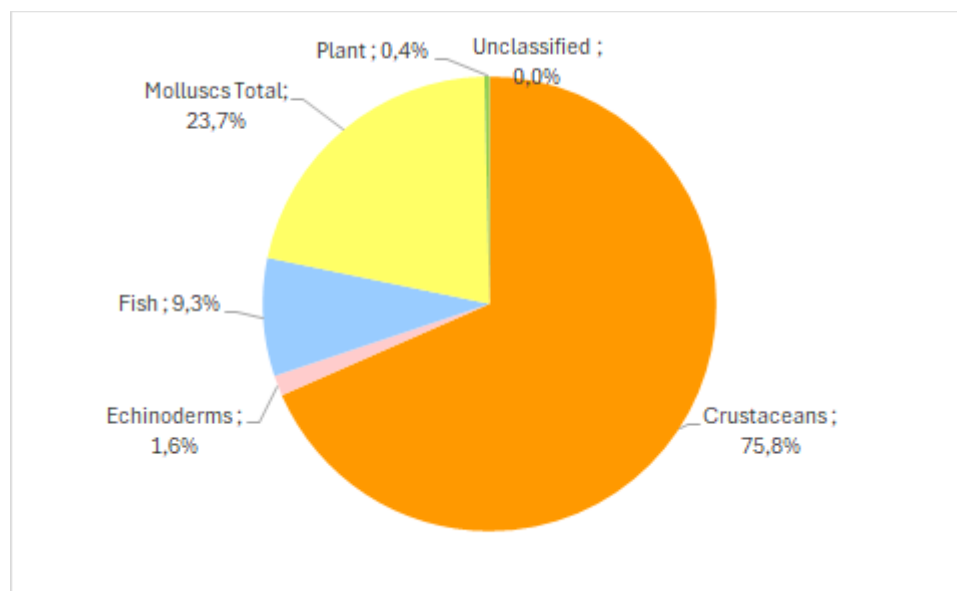


Figura 17. Index of relative importance in percent (%IRI) for the general diet.

To determine the overall importance of prey groups, the index of relative importance was used (Figure 17), since it combines numbers, weight, and frequency of prey.

According to the Index of Relative Importance (%IRI), crustaceans dominate the diet, contributing 75.8% of the total relative importance (Figure 17). Mollusks are the second most significant group, with 23.7%, followed by fish, which account for 9.3%. Echinoderms represent 1.6%, while plant matter and unidentified prey items contribute 0.4% and 0.0%, respectively.

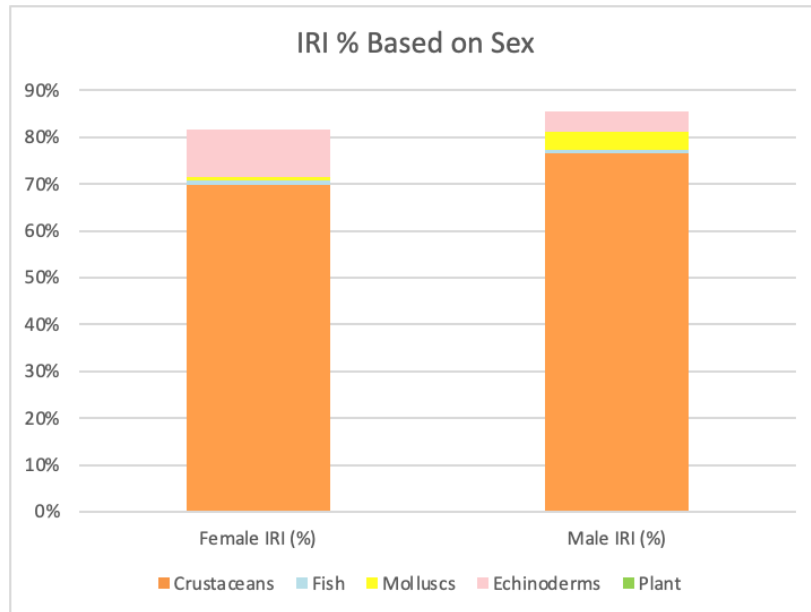


Figure 18. Percentage of Index of Relative Importance (IRI) by sex for different taxonomic groups consumed by a species, showing comparisons between female and male IRI values.

The figure shows the percentage of the Index of Relative Importance (IRI%) for different taxonomic groups consumed by males and females of a species, highlighting both similarities and differences in their diets. Crustaceans dominate the diet for both sexes, accounting for 70% of the female IRI and 77% of the male IRI, indicating their critical ecological importance as a primary prey item. Other prey groups, including fish, mollusks, and echinoderms, make only a small contribution to the diet. For females, fish and mollusks each account for just 1% of their diet. In males, mollusks make up a slightly higher portion (4%), while echinoderms are less common (4%) compared to females, where they account for 10%. Plants are completely absent from the diet of females and make up only about 1% of the diet in males. These patterns suggest dietary specialization on crustaceans, with minor sex-based differences.

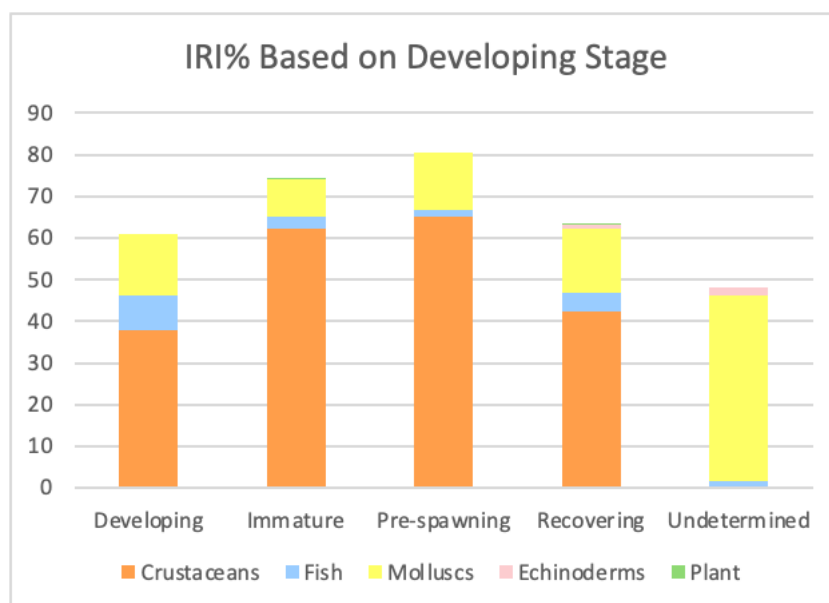


Figure 19. Percentage of Index of Relative Importance (IRI) for different taxonomic groups across developmental stages of a species, showing changes in dietary composition during different life history phases.

Figure 19 illustrates the percentage of the Index of Relative Importance (IRI%) for different taxonomic groups consumed by a species across developmental stages, highlighting changes in dietary composition. Crustaceans dominate the diet in all stages, particularly in the pre-spawning (65%) and immature (62.37%) stages, underscoring their importance as a primary prey source. In contrast, molluscs show a notable increase in importance in the recovering (15.15%) stage, indicating a shift toward greater dietary diversity in later life stages. Fish contribute minimally to the diet, with their highest proportion in the developing stage (8.22%), while echinoderms and plants are negligible across all stages.

4. Discussion:

4.1 Age-Growth

The lack of a statistically significant difference in sex distribution across length categories suggests that observed variations between males and females are likely due to random chance or an insufficient sample size. The similar patterns across most length classes indicate that the sampling effort was unbiased with respect to sex. Higher counts in certain length classes, such as 18 cm, 26 cm, and 31 cm, may reflect dominant size cohorts or preferred sizes linked to

growth or ecological factors. Meanwhile, gaps in other classes, such as 11 cm, highlight potential sampling biases or natural variability in size structure.

The age-length key results reveal distinct growth dynamics between sexes, offering valuable insights into their life history traits. Males exhibit a higher growth rate and earlier attainment of larger size classes, potentially linked to reproductive or territorial advantages. In contrast, females show stronger representation in older age groups, suggesting greater longevity tied to reproductive strategies and energy allocation. The high R^2 values for both sexes confirm the robustness of the age-length relationship, reflecting consistent and predictable growth patterns that inform sex-specific ecological roles.

The observed growth patterns of *H. didactylus* indicate slight sexual dimorphism, with females predicted to grow larger while males exhibit faster initial growth. Maximum observed sizes (37 cm for females and 39 cm for males) fall short of predicted asymptotic lengths, suggesting that environmental pressures or mortality prevent individuals from reaching their theoretical maximum sizes. These discrepancies could reflect ecological or fishing pressures, where individuals may not survive to full maturity (Palazón-Fernandez et al., 2009; Froese & Binohlan, 2000).

Males' higher growth coefficient ($K = 0.147$) compared to females ($K = 0.143$) supports the expectation of earlier maturity and faster initial growth, likely driven by reproductive strategies such as securing mates or establishing territories (Jørgensen et al., 2005; Roff, 2002). This sexual dimorphism reflects a trade-off, with males prioritizing rapid growth for early maturity (around 2 years) and females investing in longevity and long-term growth (Hutchings, 1993; Palazón-Fernandez et al., 2009).

Finally, the higher natural mortality rate in females may reflect different ecological pressures influencing growth and survival dynamics. These findings underscore the importance of sex-specific management strategies to address distinct life-history strategies, ensuring sustainable fisheries and conservation outcomes (Wootton, 1998; Roff, 2002).

4.2 Reproduction

The distribution of maturity stages provides valuable insights into the reproductive dynamics of the population. The prevalence of the Recovering stage among both sexes suggests that sampling occurred primarily post-spawning, as most individuals had already completed

reproduction. The low proportions of Pre-Spawning and Spawning stages further support this conclusion, indicating a reproductive season influenced by environmental factors (Flores et al., 2019; Modesto & Canário, 2003). For undetermined individuals, the dominance of the Immature stage reflects the presence of juveniles whose gonads were insufficiently developed for sex determination. Additionally, the inclusion of individuals in the Uncertain stage highlights the challenges of macroscopic assessments of maturity, particularly for younger or less-developed specimens (Aissat-Ziamni et al., 2020).

Patterns in gonad and liver weights illustrate the interconnected roles of reproductive investment and energy allocation. Gonad weights reveal sex-specific reproductive strategies, with females exhibiting higher gonad weights than males, consistent with the greater energetic demands of egg production (Costa et al., 2000). Males, on the other hand, show lower gonad weights, reflecting the reduced energy required for sperm production. Across maturity stages, gonad weights peaked in Spawning individuals, representing maximum reproductive effort, and were elevated in Pre-Spawning individuals preparing for reproduction. In contrast, lower gonad weights were observed in Immature, Developing, Recovering, and Uncertain stages, reflecting reproductive inactivity, ongoing maturation, or recovery post-spawning (Flores et al., 2019). Pairwise comparisons of gonad weights showed significant increases between Immature and Pre-Spawning stages ($p = 0.0062$), aligning with physiological changes associated with reproductive readiness. Significant differences between Pre-Spawning and Recovering ($p = 0.0053$) suggest shifts in energy allocation during these transitions. However, the lack of significant differences between Developing, Spawning, and Uncertain stages implies that gonad weights stabilize or follow similar patterns once reproductive activity begins, with other factors potentially influencing energy demands during these phases.

Liver weight patterns further emphasize the role of maturity stages in energy allocation. Liver weights remained consistently low during Immature, Developing, and Pre-Spawning stages, reflecting prioritization of energy for growth and gonad development. Conversely, liver weights increased during the Recovering stage, suggesting energy reserves were being replenished following the metabolic costs of reproduction. The Kruskal-Wallis rank sum test showed no significant differences in liver weight across maturity stages ($p = 0.07394$), although the p-value was close to the significance threshold, hinting at potential trends that may require larger sample sizes to confirm. Variability in liver weights within maturity stages or confounding factors could also obscure these trends, underscoring the need for more robust sampling or refined analyses to clarify these patterns (Palazón-Fernández et al., 2001).

The contrasting trends in gonad and liver weights highlight a trade-off in energy allocation. During stages of peak reproductive effort, such as Spawning and Pre-Spawning, energy is directed toward gonad development, leading to relatively low liver weights. During recovery, the liver plays a central role in replenishing energy stores, while gonad activity decreases. This interplay underscores the dynamic nature of energy use during the reproductive cycle (Hyslop, 1980).

The observed trends in Gonadosomatic Index (GSI) and Hepatosomatic Index (HSI) further illustrate the distinct reproductive and energy allocation strategies between sexes. For males, the highest GSI during the Developing stage (II) reflects peak gonadal development and active reproductive investment, while low GSI in the Immature stage (I) indicates minimal reproductive activity. For females, GSI peaks during the Pre-Spawning stage (III), underscoring significant energetic investment in egg production (Modesto & Canário, 2003; Palazón-Fernández et al., 2001). Lower GSI values in Immature (I) and Recovering (V) stages highlight periods of reduced reproductive effort or post-spawning recovery. The higher GSI values in females overall reaffirm their greater reproductive investment compared to males (Félix et al., 2015).

HSI trends complement these findings, revealing sex-specific patterns of energy storage and use. In males, the highest HSI during the Recovering stage (V) suggests replenishment of energy reserves post-spawning, while the lowest HSI in the Developing stage (II) reflects mobilization of energy for reproduction (Silva et al., 2000). In females, the highest HSI during the Immature stage (I) likely represents the buildup of energy reserves for future reproduction, while the lowest HSI in the Pre-Spawning stage (III) reflects substantial energy expenditure for gamete production. These trends reveal greater variability in HSI among females, driven by the metabolic demands of reproduction (Modesto & Canário, 2003).

The biological implications of these findings highlight distinct sex-specific reproductive strategies. Females prioritize building and utilizing energy reserves for egg production, while males maintain steadier energy use and focus on recovery post-spawning. These strategies align with observations in other toadfish species, where females demonstrate higher reproductive investment, and males emphasize sustained reproductive activity (Palazón-Fernández et al., 2001; Félix et al., 2015). Fisher's Exact Test showed no significant association between sex and maturity ($p = 0.1123$), suggesting that maturity stage distribution does not differ significantly by sex in this dataset. However, the close-to-threshold p-value, combined with the potential effects of sample size and variability, suggests that further research with larger

and more balanced datasets is warranted to clarify potential sex-maturity relationships (Flores et al., 2019).

4.3 Diet

The dietary analysis reveals a strong preference for crustaceans, particularly brachyuran crabs, which dominate the diet of *H. didactylus*. This reliance underscores the ecological importance of crustaceans, likely due to their abundance, accessibility, and high nutritional value in the Ria Formosa environment (Costa et al., 2000; Félix et al., 2015). The dominance of brachyuran crabs highlights the species' specialized foraging behavior, as these preys are consistently targeted across habitats. Crustaceans were the primary dietary component across all indices analyzed, indicating their significance both in terms of frequency and biomass. This strong reliance reflects the predator's benthic feeding behavior and preference for habitats where crustaceans are readily available (Silva et al., 2000). These findings emphasize the consistent ecological role of crustaceans as a critical prey group for *H. didactylus*.

Mollusks, primarily gastropods, represented the second most significant prey group, with their high frequency of occurrence (50.6%) indicating their availability as a reliable alternative food resource. Mollusks likely play a supplementary role in the diet, being smaller in size and contributing less in terms of biomass than crustaceans. Their presence in the diet suggests an opportunistic feeding strategy that allows the predator to exploit secondary prey groups when crustacean availability is reduced (Costa et al., 2003). This adaptability may be particularly important in dynamic estuarine environments where prey availability fluctuates seasonally or spatially (Hyslop, 1980).

Fish, although less frequent (27.0%), made a notable contribution to the diet in terms of biomass (22.6%), suggesting their role as an occasional but energetically valuable prey group. This pattern implies that *H. didactylus* may opportunistically target fish when larger prey are available, maximizing the energetic return per foraging effort. This dietary flexibility reflects an ability to exploit high-value prey during periods of abundance while maintaining a consistent reliance on crustaceans as the primary resource (Félix et al., 2015). Fish consumption could also be influenced by life stage or specific environmental factors that increase the accessibility of this prey group (Pinkas et al., 1971).

Echinoderms, plants, and unclassified prey items displayed minimal contributions to the diet. The low indices associated with echinoderms likely reflect their harder structures and reduced nutritional value, making them less favorable as prey. Similarly, plants and unclassified items

appear to be incidentally ingested rather than actively targeted, as their presence likely results from the predator's benthic foraging behavior. These components are therefore not meaningful dietary items and do not significantly influence the overall feeding strategy of the species (Hyslop, 1980).

The observed dietary variability, particularly in the contributions of mollusks and fish, may reflect seasonal or habitat-specific differences in prey availability. For instance, increased mollusk consumption during certain life stages or environmental conditions suggests shifts in dietary composition based on resource availability. This adaptability in prey selection likely contributes to the species' ecological success, allowing it to thrive in diverse estuarine and coastal lagoon habitats (Costa et al., 2003). The ability to exploit both dominant and supplementary prey groups highlights a flexible feeding strategy tailored to varying environmental conditions.

The consistent dominance of crustaceans across numerical, gravimetric, and %IRI indices further underscores their critical role in the diet. Mollusks and fish, while secondary, provide complementary resources that enhance dietary diversity. These findings align with previous studies, emphasizing the trophic role of *H. didactylus* as a benthic predator with specialized feeding strategies targeting crustaceans while maintaining flexibility for opportunistic feeding (Félix et al., 2015; Silva et al., 2000). The predator's reliance on crustaceans, complemented by the occasional inclusion of mollusks and fish, reflects a balance between opportunism and selectivity, shaped by ecological factors such as habitat and prey availability.

4.4 Limitations

This study's main limitation is its cross-sectional design, which captures data at a single point, restricting causal inferences and overlooking temporal patterns. For instance, changes in gonad and liver weights across maturity stages may not reflect long-term trends or seasonal variations. A longitudinal design would allow tracking of fluctuations over time, offering insights into consistent or transient patterns.

Refining maturity classifications could improve the accuracy of analyses and deepen understanding of reproductive strategies. Additionally, the sample size may be insufficient to detect subtle differences, especially between sexes and maturity stages. Expanding the sample size and ensuring balanced representation across habitats and geographic locations would enhance robustness and generalizability.

Environmental factors like temperature, food availability, and pollutants, which influence physiological traits, were not considered. Including these variables would provide a more holistic view of the factors shaping biological measurements. Future studies should also investigate genetic factors and underlying biological mechanisms to better understand life history strategies and inform conservation efforts.

The quality of the raw data on reproduction represents a potential limitation in this study, as the limited sample size and exclusion of data due to inaccuracies may have introduced biases into the analysis. The absence of certain prey groups in specific samples could be due to natural variability in prey availability or gaps in sampling efforts. To obtain a more representative understanding of the dietary and reproductive aspects of *H. didactylus*, increasing the sample size and ensuring a balanced sex ratio in future studies would enhance the precision of the findings. This would provide deeper insights into the feeding ecology and life-history strategies of this species, particularly in relation to its adaptation to dynamic estuarine systems.

5. Conclusion

This study provides a comprehensive evaluation of the biological and ecological traits of *H. didactylus* in Ria Formosa. Age and growth analyses confirmed sex-specific growth dynamics, with females exhibiting a higher asymptotic length and males maturing faster. Reproductive patterns revealed critical insights into energy allocation strategies, highlighting the interplay between gonad development and liver reserves across maturity stages. Dietary assessments emphasized the species' reliance on crustaceans, particularly brachyuran crabs, as a primary prey group, with secondary contributions from mollusks and fish. These findings underscore the importance of *H. didactylus* as a bioindicator for monitoring ecosystem health in coastal lagoons. Future research should explore long-term population dynamics, environmental stressors, and the implications of anthropogenic impacts on this species and its habitat to inform effective management and conservation measures.

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